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Duality, Gale Diagrams, and Applications

More about life in high dimensions: after “successfully” dealing with polytopes in four dimensions, we now study polytopes with few vertices, that is, d -polytopes with only d -plus-a-few vertices. For this, we develop a duality theory, which describes them in terms of structures in low dimensions.

This duality theory (developed by Perles in the 1960s, and recorded by Grünbaum [252]) is classically known as *Gale diagrams*. Later it was realized (apparently first by McMullen [394]) that Gale diagrams are a manifestation of “oriented matroid duality.”

Thus behind the construction of Gale diagrams one finds (barely hidden) *oriented matroids*. Their theory was initiated in the 1970s by at least four independent authors, Jon Folkman, Jim Lawrence, Robert G. Bland, and Michel Las Vergnas; see [207] and [100].

By now oriented matroids form a theory with many facets, extensive enough to fill thick books [96]. One aim of our lectures is to give a simple introduction to a few topics that help understand polytopes. Keys to this include the description of the relative position of vertices in terms of “circuits” and “cocircuits,” as well as the simple duality between these two descriptions, *oriented matroid duality*, which in polytope theory manifests itself in (linear and affine) Gale diagrams.

Thus this lecture includes a brief crash (?) course on oriented matroids. (This will be continued in Lecture 7, when we discuss hyperplane arrangements and zonotopes.) Although oriented matroids need some amount of new notation and terminology, there is little magic involved: just don’t be scared of names. For further reading, we refer to the “Orientation Session” in Björner et al. [96, Ch. 1].

In this lecture, we give several striking applications of Gale diagrams and oriented matroid duality, among them

- the construction of an 8-polytope with 12 vertices that is nonrational, meaning that it cannot be realized with rational vertex coordinates (due to Perles),
- the construction of a 5-polytope for which the shape of a 2-dimensional face cannot be prescribed (new!), and
- the construction of a 24-polytope with 28 vertices for which the realization space is not connected.

These examples are “easily” derived from special low-dimensional point configurations via Gale diagrams.

Finally, we develop the “Lawrence construction,” a systematic method by which properties of arbitrary point configurations can be encoded into polytopes. This makes it possible to provide the corresponding “universality theorems,” which say that the realization spaces of convex polytopes can be arbitrarily bad, in a sense that we make precise ahead.

6.1 Circuits and Cocircuits

For this section, let $X = \{\mathbf{x}_1, \dots, \mathbf{x}_n\} \subseteq \mathbb{R}^d$ be a finite set of n points in affine space $\mathbb{R}^d$ — for example, the vertices of a d -polytope. There is no problem with multiple points, but we always assume that the points in X affinely span $\mathbb{R}^d$. We continue our habit to interpret X as a matrix $X \in \mathbb{R}^{d \times n}$ when this is convenient.

We now explore two “dual” ways to derive combinatorial data from such a point configuration.

(a) Affine Dependences

The *affine dependences* of the point configuration X are the vectors $\mathbf{z} \in \mathbb{R}^n$ with $\mathbb{1}\mathbf{z} = 0$ such that $X\mathbf{z} = \mathbf{0}$. These vectors $\mathbf{z}$ form a vector subspace of $\mathbb{R}^n$,

$$\text{a-Dep}(X) := \{\mathbf{z} \in \mathbb{R}^n : X\mathbf{z} = \mathbf{0}, \mathbb{1}\mathbf{z} = 0\}.$$

Geometrically, let $\mathbf{z} \neq \mathbf{0}$ be an affine dependence. Then we can look at the sets of negative and positive coefficients of $\mathbf{z}$, $N(\mathbf{z}) := \{i : z_i < 0\}$ and $P(\mathbf{z}) := \{i : z_i > 0\}$. We get that $\Lambda := \sum_{i \in P(\mathbf{z})} z_i = -\sum_{i \in N(\mathbf{z})} z_i > 0$. By multiplying with $\frac{1}{\Lambda}$ we can rewrite the affine dependence $\mathbf{z}$ as

$$\sum_{i \in P(\mathbf{z})} \frac{z_i}{\Lambda} \mathbf{x}_i = \sum_{i \in N(\mathbf{z})} \frac{-z_i}{\Lambda} \mathbf{x}_i =: \mathbf{y},$$